

Registrar Desk - Simple Plan & Solution

E-zentra | Mati School of Arts and Trades For: Registrar, school staff, and anyone non-technical Date: October 5, 2026 Scope: Registrar pages on the website (Overview, Accounts, Final Grades, SF10, Adviser Access, Subjects & Sections)

TL;DR (read this in 30 seconds)

The registrar pages will soon update by themselves - no refresh needed.

Approve a student and they disappear from Pending instantly.

When something needs you (new sign-up, adviser request, finals ready, SF10 to validate), a clear pop-up appears in about 5 seconds and the bell count goes up.

Every list shows 15 rows per page, the same everywhere.

Messages are specific, never just "New notification".

Example: instead of "New notification", you will see "New G11 sign-up: Juan Dela Cruz - awaiting approval."

1. The problems we are fixing

#	What you feel today	Why it happens
1	You have to refresh to see new data.	Registrar pages have no live listener yet. Other desks (teacher. nurse.
2	Approve feels slow - the row stays until reload.	The screen waits for the server before updating.
3	Pages show different counts (8 here, 20 there. 100 loaded at once).	No standard page size. Some pages load everything then slice it.
4	Pop-ups say "New notification" - not helpful.	No registrar-specific message titles yet.
5	Sometimes you see old data after someone else changes something.	Memory (cache) is cleared too broadly or not at the right time.
6	You are not notified when advisers request SF10 access or finals become ready.	The system notifies the adviser, but not the registrar in reverse.
		Nothing is broken beyond repair - the building registrar desk.

2. What you will notice after the

Instant buttons. Tap Approve / Reject / Validate undoes itself and tells you why.

Live pages. Open two windows: approve in one refresh.

Bell that works. The bell icon in the top bar shows the rest. Tapping an item takes you to the right.

Always 15 per page. Accounts, Final Grades, your place when you go Back.

Clear pop-ups (sileo). Each pop-up names what

3. The solution in plain language

A. Live updates (no refresh)

Think of it like a messenger: the registrar screen "listens" for new inbox letters every 5 seconds, plus a bonus instant channel when available.

First check only remembers what you already saw (so no flood of old pop-ups).

New letters after that pop up (max 3 at a time, oldest first).

Every pop-up also refreshes the right list quietly in the background.

When you return to the tab, it checks immediately.

Technical note for developers: clone of teacherChannel.ts - 5s auth-gated GET /api/notifications/ poll is the real transport (Supabase anon Realtime cannot see backend-JWT rows), Supabase channel is bonus, seenIds dedup, 2-second invalidate throttle, refetch on focus.

B. Fast buttons (optimistic UI)

You tap -> screen updates instantly (row removed / badge flips).

App sends the request in the background.

If success -> small success message + lists re-check quietly.

If fail -> change is undone + clear error message.

Your own tap never causes a duplicate pop-up - the system suppresses its own echo for 30 seconds but still updates lists and the bell.

C. Smart memory (caching, 2 layers)

Server memory (Redis, 30 seconds): Going Back feels instant. Keyed by page + filters + your role + school year/term, so teachers never see each other's data and Term 1 never shows as Term 2.

Screen memory (React Query, 30 seconds): Moving between tabs shows cached data instantly, no flashing.

Correct clearing: Every Approve / Validate / Release clears only the affected memory (e.g. approve student clears Accounts + Overview, not everything).

New narrow tags: registrar-overview, registrar-finals, registrar-accounts, registrar-sf10, registrar-access, registrar-academics.

D. 15 per page, everywhere

Default pageSize = 15, server-paginated. Overview cards show first 5 + "View all" link to the full 15-paged page.

Page turns keep old data on screen while loading the next page (no skeleton flash).

E. Notifications that tell you exactly what happened

The system will now notify the registrar (not just the adviser) when:

Event	Who tells registrar	Example message
New G11/G12 student signs up	Sign-up form	New G11 sign-up: Juan Dela Cruz (LRN 123456789012) - awaiting
Adviser requests SF10 access	Adviser request form	SF10 access requested: Adviser Ana Reyes - 11-Bayanihan (23 advisees).
All subjects adviser-approved for a student-term	Grade pipeline	Finals ready: Maria Santos (11-Bayanihan) - Term 1. all subjects
SF10 verified, needs validation	Teacher verify	SF10 verified: Ana Lim (G12) - needs validation.
SF10 released / validated	SF10 flow	SF10 released: Jose Rizal (G11) - ready.

Pop-up titles match bell titles exactly (one shared mapper), derived from the message - never generic. Examples of titles: "New student sign-up", "SF10 access requested", "Finals ready for review", "SF10 needs validation", "Academics updated".

4. What stays safe

Grade band: Registrar only ever sees G11-G12. Record Keeper sees G7-G10. Same code, scoped by role.

View-only finals: Registrar never approves grades - only views fully adviser-approved sets. That rule does not change.

Audit trail: Every Approve / Validate / Release / Create still writes an audit row (who, what, when, why).

Confidentiality: No clinical write-ups are exposed. Principal status-only rule unchanged.

No spam: Same event within 60 seconds with identical message never creates a second inbox row. Parallel saves are locked so double-clicks do not double-create.

5. Work plan (in order)

Backend tells registrar - add 4 fan-outs (sign-up, access-request created, finals-ready once per student-term, SF10 verified) + message templates + notification types.

Live listener + bell - new registrarChannel.ts, mount in registrar layout, add bell to top bar with deep links. 15 per page - fix Accounts (8 -> 15), Final Grades (100-fetch/20-slice -> server page 15), SF10, Access, Audit; cap Overview previews to 5.

Instant buttons - optimistic Approve/Reject, Access Approve/Deny, SF10 Validate/Release, Subject/Section/Assignment create/edit/delete with rollback.

Narrow cache clearing - replace broad registrar tag with per-resource tags.

Test - two-window live check (tap in one, see it in the other in ~5s, no refresh) + automated tests for fan-out, dedup, paging + check x-cache: HIT/MISS.

Appendix A - For developers (files to touch)

Backend

src/lib/notify.ts - add TYPE_MAP: users:pending_signup -> account_pending, adviser_sf10_access_requests:create -> sf10_access_requested, final_grades:adviser_approved -> finals_ready, sf10_records:verified -> sf10_needs_validation. Use fanoutToRole("registrar", { messageFor }), void after res.json.

src/modules/auth/auth.routes.ts - fan-out on G11-G12 student sign-up; add page/pageSize to pending list.

src/modules/registrar/registrar.routes.ts - clamp final-grades pageSize max 15; accounts-audit default 15; overview latestAttachments take 100 -> 5, cap missingSf10/pendingStudents previews to 5; narrow cache() tags + invalidateTags() per write.

src/modules/registrar/academics.routes.ts - narrow tags; keep band guard GRADE_BAND = [G11, G12].

src/modules/sf10/sf10.routes.ts - fan-out on verify (to registrar/record_keeper by band); narrow invalidates on validate/release.

Frontend

NEW src/lib/realtime/registrarChannel.ts - mirror teacherChannel.ts structure; REGISTRAR_KEYS = [registrar-overview, registrar-final-grades, pending-students, account-breakdown, accounts-audit, registrar-sf10, registrar-access, registrar-notifications]; FALLBACK_POLL_MS = 5000, MAX_TOASTS_PER_POLL = 3, markSelfNotified() + 30s suppress.

src/app/registrar/layout.tsx - mount useRegistrarRealtime() + add NotificationsBell (queryKey=["registrar-notifications"], resolveTarget: pending -> /registrar/accounts, finals -> /registrar/final-grades, access -> /registrar/adviser-access, SF10 -> /registrar/sf10).

src/lib/notifications/label.ts - add shared registrar title mapper so bell and sileo match.

src/app/registrar/accounts/page.tsx - PAGE_SIZE 8 -> 15, queryKey ["pending-students", page], server page param, placeholderData: keepPreviousData, optimistic approve/reject.

src/app/registrar/final-grades/page.tsx - send page + pageSize: 15 (stop fetching 100), queryKey includes page, remove client 20-slice.

src/app/registrar/sf10/*, adviser-access/*, AccountsAudit - same 15 pattern + optimistic validate/release/approve/deny.

src/components/providers.tsx - no change needed (staleTime 30s already correct).

Verify: vitest (fan-out, 60s dedup, paging clamp), manual two-tab check, x-cache header check, bell deep-link check.

Appendix B - Copy-paste build prompt for AI

Implement the Registrar realtime plan in E-zentra. Constraints: strict pageSize 15 everywhere (server-paginated: registrar/final-grades clamp max 15, accounts-audit default 15, auth/pending add page/pageSize, overview previews take 5); narrow backend cache tags (registrar-overview/finals/accounts/sf10/access/academics) with matching invalidateTags on writes; new frontend lib/realtime/registrarChannel.ts cloning teacherChannel.ts (5s auth poll + Supabase INSERT channel, seenIds dedup, max 3 toasts/poll, 2s invalidate throttle, focus refetch) mounted in app/registrar/layout.tsx plus NotificationsBell with role-scoped resolveTarget; backend fan-outs TO registrar (signup, access-request create, finals-ready once per student-term, SF10 verified) with specific messages via fanoutToRole + messageFor, TYPE_MAP entries, void-after-response, 60s dedup respected; specific-always sileo/bell titles via shared toastTitleForRegistrar (message-regex, never generic); optimistic TanStack mutations (onMutate setQueryData + rollback + markSelfNotified self-suppress + onSettled narrow invalidates) for approve/reject, access approve/deny, SF10 validate/release, subject/section/assignment CRUD. Keep grade-band scoping (registrar G11-G12), termScope in keys, existing design system. Verify with vitest + two-tab live check, no manual refresh.

End of report. Questions? Start with Section 2 (what you will notice) - everything else is how we get there.